LAB | Query Optimization, Indexing, Storage Performance, and Data Formats

Week 2 — Day 5: Query Optimization, Indexing, Storage Performance, and Data Formats — Lab

Practice Workbook

BritMart has working queries, but some of them read more data than needed. The team also needs to choose sensitive storage layouts and file formats for analytical data.

This lab uses MySQL skills already covered in Week 1 and the performance and file-format concepts covered today. It does not require tools that have not yet been introduced.

Part 1 — Query Performance, Full Scans, and Indexes

Q. 1 — Create a Table for Query Testing

Create:

💡 orders_perf

with:

💡 order_id
customer_id
order_date
city

```
amount_gbp
status
```

Insert these rows:

```
1, 101, 2026-08-01, London, 45.50, complete
2, 102, 2026-08-01, Manchester, 18.00, cancelled
3, 101, 2026-08-02, London, 62.25, complete
4, 103, 2026-08-02, Leeds, 30.00, complete
5, 104, 2026-08-03, Bristol, 55.50, complete
6, 102, 2026-08-03, Manchester, 12.00, pending
7, 101, 2026-08-04, London, 80.00, complete
8, 105, 2026-08-04, Leeds, 20.00, complete
9, 106, 2026-08-05, Bristol, 100.00, complete
10, 102, 2026-08-05, Manchester, 40.00, complete
```

Solution:

Q. 2 — Inspect a Query Before Adding an Index

Run:

```
EXPLAIN
SELECT
order_id,
order_date,
amount_gbp
FROM orders_perf
WHERE customer_id = 101;
```

Remember what MySQL shows for the access type and possible key.

Do not expect a fixed plan on every machine. The goal is to inspect what your database chooses.

Solution:

'-> Filter: (orders_perf.customer_id = 101) (cost=1.25 rows=1)\n   -> Table scan on orders_perf (cost=1.25 rows=10)\n'

Q. 3 — Add an Index and Compare

Create:

> 💡 idx_orders_customer

on:

> 💡 customer_id

Run the same `EXPLAIN` again.

Record:

> 💡 before index
> after index

and describe any difference.

Solution:

'-> Index lookup on orders_perf using idx_orders_customers (customer_id = 101) (cost=0.8 rows=3)\n'

Q. 4 — MCQ: Index

When is an index most likely to help?

Solution: B. A query filters a large table using a selective column.

Part 2 — Filtering Early, Selecting Columns, and Join Keys

Q. 5 — Compare `SELECT *` with Required Columns

Run both queries.

Query A

```
SELECT *
FROM orders_perf
WHERE status = 'complete';
```

Query B

```
SELECT
order_id,
order_date,
amount_gbp
FROM orders_perf
WHERE status = 'complete';
```

Explain why Query B is better when those are the only columns the consumer needs.

Solution:

Query A traverses the entire table with all the columns that is not required whereas Query B is faster as it traverses only the selected columns.

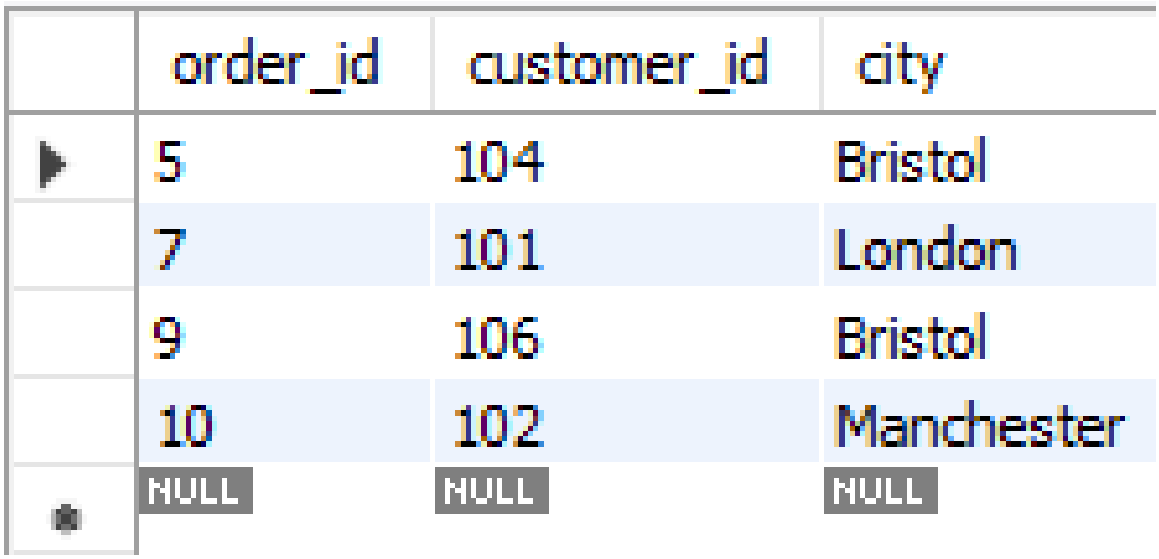
Q.6 — Filter Before Returning Data

BritMart only needs:

completed orders
from 2026-08-03 onwards
amount at least £40

Write one query that applies all three filters.

Solution:



	order_id	customer_id	city
▶	5	104	Bristol
	7	101	London
	9	106	Bristol
	10	102	Manchester
*	NULL	NULL	NULL

Q. 7 — Choose the Correct Join Key

Create:

customers_perf

with:

customer_id
customer_name

city

Insert:

101, Amelia Clarke, London
102, Oliver Bennett, Manchester
103, Aisha Khan, Leeds
104, Harry Thompson, Bristol
105, Emily Wilson, Leeds
106, Noah Davies, Bristol

Join customers to orders using the correct key.

Explain why joining on `city` would be unsafe.

Solution:

	order_id	customer_id	order_date	amount_gbp
▶	1	101	2026-08-01	45.50
	3	101	2026-08-02	62.25
	7	101	2026-08-04	80.00
	2	102	2026-08-01	18.00
	6	102	2026-08-03	12.00
	10	102	2026-08-05	40.00
	4	103	2026-08-02	30.00
	5	104	2026-08-03	55.50
	8	105	2026-08-04	20.00
	9	106	2026-08-05	100.00

It would be risky to join both tables based on columns because they are not PRIMARY and FOREIGN keys.

Q. 8 — MCQ: Required Columns

Why should a query select only the columns it needs?

solution: C. It can reduce unnecessary data read or transferred

Part 3 — Partitioning, Small Files, Columnar Storage, and Compression

Q. 9 — Choose a Partitioning Column

BritMart stores several years of order data.

Most reports filter by:



order_date

Consider these possible partition columns:



order_id
order_date
customer_name
amount_gbp

Which is the best starting choice for the stated access pattern? Explain why.

solution:

Starting choice would be order_date , range partitioning divides data into partitions based on the specified range of values, it will further help in summarizing monthly, yearly sales.

Q. 10 — Design a Date-Based Layout

BritMart has these three daily batches:

2026-08-01 → 100,000 rows
2026-08-02 → 120,000 rows
2026-08-03 → 90,000 rows

Create a simple storage layout using the date.

Solution:

```
orders/
    order_date=2026-08-01/
    order_date=2026-08-02/
    order_date=2026-08-03/
```

Q. 11 — Find the Small-Files Problem

Two designs contain the same amount of data.

Design A

100 files of about 100 MB each

Design B

100,000 files of about 0.1 MB each

Which design is more likely to create excessive file-management and task overhead? Explain why.

Solution:

Q. 12 — MCQ: Compression

What is the main storage benefit of compression?

Solution:

A. It reduces the amount of storage required for compressible data.

Part 4 — CSV, Parquet, Avro, and ORC

Q. 13 — Compare the Formats

Complete the table.

Format	Text or binary?	Row or column oriented?	Common strength
CSV	Text	Row	human readable
Parquet	Binary	Column	efficient analytical scans
Avro	Binary	Row	compressed
ORC	Binary	Column	compression and analytical storage

Q. 14 — Choose a Format for Each Requirement

Choose **CSV** , **Parquet** , **Avro** , or **ORC** .

A. A simple file that a business user can open easily.
B. Analytical data where queries often read only a few columns.
C. Record-oriented data exchange where schema travels with the data.
D. Columnar analytical storage used with systems that support ORC well.

Solution:

A. A simple file that a business user can open easily: **CSV**
B. Analytical data where queries often read only a few columns: **Parquet**
C. Record-oriented data exchange where schema travels with the data: **AVRO**
D. Columnar analytical storage used with systems that support ORC well: **ORC**

Q. 15 — Inspect a CSV and List Its Limitations

Create:

orders_sample.csv

with:

```
order_id,order_date,city,amount_gbp,status
1,2026-08-01,London,45.50,complete
2,2026-08-01,Manchester,18.00,cancelled
3,2026-08-02,Leeds,30.00,complete
```

Open it in a text editor.

Write down:

- how the schema is represented;
- whether the file is human-readable;
- two limitations compared with a binary analytical format.

Solution:

- Tabular, human-readable file.
- yes, it is.
- requires large storage.

Q. 16 — MCQ: Parquet

Why is Parquet commonly used for analytical data?

Solution:

D. Its columnar layout can help queries read only needed columns.